

Stony Brook University
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MIDTERM EXAM IV

NAME: _____

ID: _____

Instructions: Answer all questions. The total number of points for the exam is 100. The number of points assigned to each section is indicated. Each question within a section carries the same weight. Calculators are allowed but other electronic devices or outside materials are NOT allowed. For questions with a numerical answer, give a numerical answer. For example

$$2 + V[X]$$

is NOT the final answer if you can compute a numerical value for $V[X]$.

You do not need to compute probabilities using any of the distributions discussed during the lectures. This is, your answer can be something like

$$1 - \Phi(1.67)$$

and you don't have to calculate that this number is actually 0.04746.

You are **NOT** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Part I: Multiple Choice Questions (20 points)

1. A 95% confidence set for two or more coefficients is a set that contains
 - (a) the same values as the 95% confidence intervals constructed for the coefficients.
 - (b) integer values only.
 - (c) the population values of these coefficients in 95% of randomly drawn samples.
 - (d) the sample values of these coefficients in 95% of randomly drawn samples.
2. When there are two coefficients, the resulting confidence sets are
 - (a) squares
 - (b) rectangles
 - (c) ellipses
 - (d) trapezoid
3. Consider a regression with two variables, in which X_{1i} is the variable of interest and X_{2i} is the control variable. Conditional mean independence requires
 - (a) $E[u_i | X_{1i}, X_{2i}] = E[u_i | X_{2i}]$
 - (b) $E[u_i | X_{1i}, X_{2i}] = E[u_i | X_{1i}]$
 - (c) $E[u_i] = E[u_i | X_{2i}]$
 - (d) $E[u_i | X_{1i}] = E[u_i | X_{2i}]$
4. The interpretation of the slope coefficient in the model $Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$ is as follows:
 - (a) a change in X by one unit is associated with a β_1 change in Y .
 - (b) a 1% change in X is associated with a $\beta_1\%$ change in Y .
 - (c) a change in X by one unit is associated with a $\beta_1 100\%$ change in Y .
 - (d) a change in 1% is associated with a change in Y of $0.01\beta_1$.
5. For the polynomial regression model,
 - (a) you can still use OLS estimation techniques, but the t -statistics do not have an asymptotic normal distribution.
 - (b) the critical values from the normal distribution have to be changed to 1.96^2 , 1.96^3 , etc.
 - (c) you need new estimation techniques since the OLS assumptions do not apply any longer.
 - (d) the techniques for estimation and inference developed for multiple regression can be applied.
6. To test whether or not the population regression function is linear rather than a polynomial of order r ,
 - (a) use the test of $(r - 1)$ restrictions using the F -statistic.
 - (b) check whether the regression R^2 for the polynomial regression is higher than that of the linear regression.
 - (c) look at the pattern of the coefficients: if they change from positive to negative to positive, etc., then the polynomial regression should be used.

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- (d) compare the TSS from both regressions.
7. In the regression model $Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 X_i \times D_i + u_i$ where X is a continuous variable and D is a binary variable, the coefficient β_3
- (a) indicates the slope of the regression when $D_i = 1$.
 - (b) indicates the difference in the slopes of the two regressions.
 - (c) has a standard error that is not normally distributed even in large samples since D is not a normally distributed variable.
 - (d) has no meaning since $X_i \times D_i = 0$ when $D_i = 0$
8. In the model $Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 \times X_2 + u_i$, the expected effect $\frac{\Delta Y}{\Delta X_1}$ is
- (a) $\beta_1 + \beta_3 X_1$.
 - (b) $\beta_1 + \beta_3 X_2$.
 - (c) $\beta_1 + \beta_3$.
 - (d) β_1 .
9. Assume that you had estimated the following regression model

$$\widehat{TestScore} = 607.3 + 3.85Income - 0.0423Income^2.$$

If income increased from 10 to 11 (\$10,000 to \$11,000), then the predicted effect on test scores would be

- (a) 2.96
 - (b) 3.85
 - (c) $3.85 - 0.0423$
 - (d) Cannot be calculated because the function is non-linear.
10. Consider the population regression of log earnings against two binary variables: whether a worker is married $married_i$, where $married_i = 1$ if the i -th person is married; and the worker's gender $gender_i$, where $gender_i$ if the i -th person is female and an interaction term of the two binary variables. Thus, the model looks like

$$\ln(earnings)_i = \beta_0 + \beta_1 married_i + \beta_2 gender_i + \beta_3 married_i \times gender_i + u_i.$$

The interaction term:

- (a) allows the population effect on log earnings of being married to depend on gender.
- (b) cannot be estimated without the presence of a continuous variable.
- (c) does not make sense since it could be zero for married males.
- (d) indicates the effect of being married on log earnings.

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Part II: Open Problems (80 points)

Quadratic Specification and Interaction terms (30 points)

You have a data set consisting on a cross-sample of 4131 high-school students. The variables you have are: *sat*, which is the combined SAT score; *hsize* is the size of the student's high school graduating class, in hundreds; *female* is a gender dummy variable, having a value of one if female or zero if male; and *black* which is a race dummy variable equal to one for blacks and zero otherwise.

	<i>Dependent variable:</i>	
	<i>sat</i>	
	(1)	(2)
<i>hsize</i>	19.297*** (3.817)	3.933*** (1.178)
<i>hsize</i> ²	-2.195*** (0.524)	
<i>hsize</i> ² - 7 <i>hsize</i>		-2.195*** (0.524)
<i>female</i>	-45.091*** (4.230)	-45.091*** (4.230)
<i>black</i>	-169.813*** (15.434)	-169.813*** (15.434)
<i>female</i> × <i>black</i>	62.306*** (19.320)	62.306*** (19.320)
Constant	1,028.097*** (6.360)	1,028.097*** (6.360)
Observations	4,131	4,131
R ²	0.086	0.086
Adjusted R ²	0.085	0.085
Residual Std. Error	133.369	133.369
Overall F Statistic (df = 5; 4131)	77.524***	77.524***
F Statistic <i>black</i> + <i>female</i> × <i>black</i> = 0 (df=1;4131)	84.965***	
Note: *p<0.1; **p<0.05; ***p<0.01		

1. Consider the model in column (1). Write down the population model that the researcher has in mind.

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2. Is there strong evidence that $hsize^2$ should be included in the model?
3. Consider student A , which graduated from a high school class of 300 ($hsize = 3$), is female and not black. Student B graduated from a high school class of 400 ($hsize = 4$) and is also female and not black. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Rewrite the regression model so that you can construct a 95% confidence interval around Δsat . Compute this confidence interval. *Hint: You will have to use the regression in model (2). How do you rewrite the regression model so that you can use it?*
4. Holding the value of $hsize$ fixed, what is the estimated difference in SAT score between black females and nonblack females? State the null hypothesis that you would need to test to determine if the difference is statistically significant. Test it at the 5% level. After you test this null hypothesis, construct a 95% confidence interval around the estimated difference. *Hint: Look at the F statistic at the bottom of the table. What is its meaning? To construct the confidence interval, you will have to use one of the formulas we saw in class.*

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Logarithmic Specifications (30 points)

You have a data set on a cross sample of 205 firms and their chief executive officer (CEO) compensation. You have the variables $\log(\text{salary})$ which is the natural logarithm of salary (in thousands of \$); $\log(\text{sales})$ which is the natural logarithm of sales (in millions of \$); roe which is the return on equity of the firm, in percentage terms; and finance which is a dummy indicating equal to one if the firm is in the financial industry or zero otherwise. You estimate the following model.

<i>Dependent variable:</i>	
	$\log(\text{salary})$
$\log(\text{sales})$	0.274*** (0.029)
roe	0.019*** (0.003)
finance	0.183** (0.082)
Constant	4.306*** (0.258)
Observations	205
R^2	0.299
Adjusted R^2	0.289
Residual Std. Error	0.478
F Statistic	29.195*** (df = 3; 205)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

1. What is the interpretation on the coefficient on $\log(\text{sales})$? Is it statistically significant?
2. Suppose that sales increase from 20 million to 25 million, i.e. an increase in 25%, while all other variables are held constant. What is the predicted effect (in percentage terms) on salary? Construct a 90% interval around this predicted change.

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- How would you interpret the coefficient on *roe*. Holding all other factors constant, what is the percentage increase on CEO salary if return on equity increases by 10 units (10 percentage points)?
- What is the percentage difference in estimated salary between a CEO from a financial firm industry and CEO from a non-financial firm? Construct a 90% confidence interval around this percentage difference.

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Model Specification (20 points)

1. Explain why $E[u_i|X_{1,i}, \dots, X_{K,i}] = 0$ does not hold when control variables are included among the regressors.
2. Explain what is a confidence set for β_1 , β_2 , and β_3 .
3. Suppose the population model is $Y_i = f(X_i) + u_i$ where Y_i = represents grade in an econometrics course (measured from 0 to 100) for individual i and X_i a positive measure of effort of individual i . Specify (write down) a nonlinear model that you can estimate using *OLS*. Explain your choice of model.

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Formulas

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_K X_{K,i} + u_i \quad (1)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \dots - \hat{\beta}_K \bar{X}_K. \quad (2)$$

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad SSR = \sum_{i=1}^n \hat{u}_i^2. \quad (3)$$

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\sigma_u}{\sigma_X} \rho_{Xu} \quad (4)$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}. \quad (5)$$

$$\bar{R}^2 = 1 - \left(\frac{n-1}{n-K-1} \right) \frac{SSR}{TSS}. \quad (6)$$

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \quad (7)$$

$$F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right), \quad (8)$$

$$F = \frac{(R_{unrestricted}^2 - R_{restricted}^2)/q}{(1 - R_{unrestricted}^2)/(n - K_{unrestricted} - 1)} = \frac{(SSR_{restricted} - SSR_{unrestricted})/q}{SSR_{unrestricted}/(n - k_{unrestricted} - 1)}. \quad (9)$$

$$SE(\Delta \hat{Y}) = \frac{|\Delta \hat{Y}|}{\sqrt{F}} \quad (10)$$

$$90\% \text{ confidence interval for } \beta_j : \left[\hat{\beta}_j \pm 1.64 \times SE(\hat{\beta}_j) \right]. \quad (11)$$

$$95\% \text{ confidence interval for } \Delta Y : \left[\Delta \hat{Y} \pm 1.96 \times SE(\Delta \hat{Y}) \right]. \quad (12)$$

Critical Values for 2-sided t test	
Significance	Critical value
10%	1.64
5%	1.96
1%	2.58

q	5% $F_{q,\infty}$ critical value
1	3.84
2	3.00
3	2.60
4	2.37
5	2.21